18.5: Cancer results from genetic changes that affect cell cycle control

In Concept 12.3, we considered cancer as a disease in which cells escape from their normal cell division control mechanisms. Now that we have discussed the molecular basis of gene expression and its regulation, we can look at cancer more closely. The gene regulation systems that go wrong during cancer turn out to be the very same systems that play important roles in embryonic development, the immune response, and many other biological processes. Thus, research into the molecular basis of cancer has both benefited from and informed many other fields of biology.

Types of Genes Associated with Cancer

The genes that normally regulate cell growth and division during the cell cycle include genes for growth factors, their receptors, and the intracellular molecules of signaling pathways. (To review cell signaling, see Concept 11.2; for regulation of the cell cycle, see Concept 12.3.) Mutations that alter any of these genes in somatic cells can lead to cancer. The agent of such change can be random spontaneous mutation. Many cancer-causing mutations likely also result from environmental influences: chemical carcinogens, including many compounds in tobacco products, X-rays and other high-energy radiation, and some viruses.

Cancer research uncovered cancer-causing genes called **oncogenes** (from the Greek *onco*, tumor) in certain types of viruses. Later, related versions of viral oncogenes were found in the genomes of humans and other animals. The normal versions of the cellular genes, called **proto-oncogenes**, code for proteins that stimulate normal cell growth and division.

How might a proto-oncogene—a gene that has an essential function in normal cells—become an oncogene, a cancer-causing gene? In general, an oncogene arises from a genetic change that leads to an increase either in the amount of the proto-oncogene's protein product or in the intrinsic activity of each protein molecule. The genetic changes that convert proto-oncogenes to oncogenes fall into four main categories **(Figure 18.23)**: epigenetic changes, translocations, gene amplification, and point mutations.

Figure 18.23

Genetic changes that can turn proto-oncogenes into oncogenes.

Genetic changes that can turn proto-oncogenes into oncogenes.

Parts are as follows.

Part a. Epigenetic changes. Mutation in a gene for a chromatin-modifying enzyme can lead to loosening of the chromatin and inappropriate expression of a proto-oncogene.

Part b. Translocation of transposition: gene moved to new locus, or under new controls. The proto-oncogene goes to D N A with new promoter and oncogene, producing excess of normal growth-stimulating protein.

Part c. Gene amplification: multiple copies of the gene. Proto-oncogene goes to D N A with multiple oncogenes, producing excess of normal growth-stimulating protein.

Part d. Point mutation: occurs either within a control element or within the gene. Within control element: proto-oncogene goes to oncogene with mutation in one control element upstream of an oncogene. Produces normal growth stimulating protein in excess. Within the gene: Proto-oncogene goes to oncogene with a mutation within. Produces hyperactive or degradation-resistant protein.

- **Epigenetic changes that alter chromatin form.** Alterations in epigenetic modifications that can lead to abnormal chromatin condensation in a cell are often found in tumor cells. If a mutation in a gene for a chromatin-modifying enzyme leads to loosening of chromatin in a region that is normally not being expressed, a proto-oncogene in that region could be expressed at abnormally high levels (see Figure 18.23a). For example, a gene for one such enzyme has been shown to be mutated in 20% of tumor cells analyzed.
- **Movement of DNA within the genome.** Cancer cells are frequently found to contain chromosomes that have broken and rejoined incorrectly, translocating fragments from one chromosome to another (see Figure 15.14). If a translocated proto-oncogene ends up near an especially active promoter (or other control element), its transcription may increase, making it an oncogene, as shown in Figure 18.23b. (A process called transposition, which we will discuss in Concept 21.4, can also move DNA fragments.)
- **Amplification of a proto-oncogene.** Amplification increases the number of copies of the proto-oncogene in the cell through repeated gene duplication (discussed in Concept 21.5; see Figure 18.23c).
- **Point mutations in a control element or in the proto-oncogene itself.** A point mutation either in the promoter or an enhancer that controls a proto-oncogene could cause an increase in its expression. A point mutation in the coding sequence of the proto-oncogene could change the gene's product to a protein that is more active or more resistant to degradation than the normal protein (see Figure 18.23d).

Any of these four mechanisms can lead to abnormal stimulation of the cell cycle and put the cell on the path to becoming a cancer cell.

In addition to genes whose products normally promote cell division, cells contain genes whose normal products *inhibit* cell division. Such genes are called **tumor-suppressor genes** since the proteins they encode help prevent uncontrolled cell growth. Any mutation that decreases the normal activity of a tumor-suppressor protein may contribute to the onset of cancer, in effect stimulating growth through the absence of suppression.

The protein products of tumor-suppressor genes have various functions. Some repair damaged DNA, which prevents the cell from accumulating cancer-causing mutations. Other tumor-suppressor proteins control adhesion of cells to each other or to the extracellular matrix; proper cell anchorage is crucial in normal tissues and is often absent in cancers. Still other tumor-suppressor proteins are components of cell-signaling pathways that inhibit the cell cycle.

Interference with Normal Cell-Signaling Pathways

Let's consider how protein components of cell-signaling pathways function in normal cells and what goes wrong with their function in cancer cells. We will focus on the products of two key genes, the *ras* proto-oncogene and the p53 tumor-suppressor gene. Mutations in *ras* occur in about 30% of human cancers and mutations in p53 in more than 50%.

The Ras protein, encoded by the ***ras*gene** (named for rat sarcoma, a connective tissue cancer), is a G protein that relays a signal from a growth factor receptor on the plasma membrane to a cascade of protein kinases (see Figures 11.8 and 11.10). The cellular response at the end of the pathway is the synthesis of a protein that stimulates the cell cycle **(Figure 18.24a)**. Normally, such a pathway will not operate unless triggered by the appropriate growth factor. But certain mutations in the *ras* gene can lead to production of a hyperactive Ras protein that triggers the kinase cascade even in the absence of growth factor, resulting in increased cell division **(Figure 18.24b)**. In fact, hyperactive versions or excess amounts of any of the pathway's components can have the same outcome: excessive cell division.

Figure 18.24

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Normal and mutant cell cycle–stimulating pathway.
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Normal and mutant cell cycle-stimulating pathways.

Normal cell-cycle stimulating pathway. 1 growth factor (outside cell) binds to 2 its receptor in the plasma membrane. Receptor drawn with 6 bound phosphate groups. The signal is relayed to 3 a G protein, called Ras. Like all G proteins, Ras is active when G T P is bound to it (here, G T P is bound). Ras passes the signal to 4 a series of protein kinases (in the cytoplasm). The last kinase activates 5 a transcription factor

(activator) within the nucleus, shown on the D N A, that turns on one or more genes for 6 a protein that stimulates the cell cycle. Protein drawn in the cytoplasm inducing normal cell division. Mutant cell cycle-stimulating pathway. If a mutation makes Ras or any other pathway component abnormally active, excessive cell division and cancer may result. Here, a hyperactive Ras protein in the cytoplasm (product of oncogene) issues signals on its own, whether or not growth factor is bound to the receptor. No growth factor is bound, yet Ras signal proceeds, transcription factor is active in the nucleus, and protein is expressed all the time (overexpression), leading to increased cell division.

Figure 18.25a shows a pathway in which an intracellular signal leads to the synthesis of a protein that suppresses the cell cycle. In this case, the signal is damage to the cell's DNA, perhaps as the result of exposure to ultraviolet light. Operation of this signaling pathway blocks the cell cycle until the damage has been repaired. Otherwise, the damage might contribute to tumor formation by causing mutations or chromosomal abnormalities. Thus, the genes for the components of the pathway act as tumor-suppressor genes. The ***p53*gene**, named for the apparent molecular weight of its protein product, is a tumor-suppressor gene. The protein it encodes is a specific transcription factor that promotes the synthesis of cell cycle–inhibiting proteins. That is why a mutation that knocks out the p53 gene or in a gene required to activate the p53 protein (for example, a gene called *ATM*) can lead to excessive cell growth and cancer **(Figure 18.25b)**.

Figure 18.25

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Normal and mutant cell cycle–inhibiting pathway.
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Normal and mutant cell cycle-inhibiting pathways.

Normal cell cycle-inhibiting pathway. In the normal pathway, 1 D N A damage (here, UV light resulting in D N A damage in genome in the nucleus) is an intracellular signal that is passed via 2 protein kinases (span the nuclear envelope to the cytoplasm and back to nuclear envelope), leading to activation of 3 p 53. Activated p 53 (in nucleus) promotes 4 transcription (activated p 53 shown on the D N A) of the gene for 5 a protein that inhibits the cell cycle (drawn as a stop sign in the cytoplasm). Damaged D N A is not replicated and no cell division occurs. If the D N A damage is irreparable, then the p 53 signal leads to programmed cell death (apoptosis). Mutant cell-cycle inhibiting pathway. Mutations causing deficiencies in any pathway component can contribute to the development of cancer. Here, UV light causes damage in the D N A in the genome within the nucleus, sending a signal through the series of protein kinases. A mutation occurs, resulting in defective or missing transcription factor, such as p 53, which cannot activate transcription. The inhibitory protein is absent from the cytoplasm, and the cell cycle is not inhibited. Increased cell division results.

Explain whether a cancer-causing mutation in a tumor-suppressor gene, such as p53*, is more likely to be a recessive or a dominant mutation.*

*For suggested answer, see*Appendix A.**

The p53 gene has been called the “guardian angel of the genome.” Once the p53 protein is activated—say, by the ATM protein, a protein kinase, after DNA damage—p53 activates several other genes, such as p21. The p21 protein halts the cell cycle by binding to cyclin-dependent kinases, allowing time for the cell to repair the DNA. Researchers recently showed that p53 also activates expression of a group of miRNAs that inhibit the cell cycle. The p53 protein can also turn on genes directly involved in DNA repair. If DNA damage is irreparable, p53 activates “suicide” genes, whose protein products bring about programmed cell death (apoptosis; see Figure 11.20). Thus, p53 acts in several ways to prevent a cell from passing on mutations due to DNA damage. If mutations do accumulate and the cell survives through many divisions—as is more likely if the p53 tumor-suppressor gene is defective or missing—cancer may ensue. The many functions of p53 suggest a complex picture of regulation in normal cells, one that we do not yet fully understand.

A recent study of elephants may underscore the protective role of the p53 gene. The incidence of cancer among elephants in zoo-based studies has been estimated at about 3%, compared with closer to 30% for humans. Genome sequencing revealed that elephants have 20 copies of the p53 gene, compared to one copy in humans, other mammals, and even manatees, elephants’ closest living relatives. There are undoubtedly other underlying reasons, but the correlation between low cancer rate and extra copies of the p53 gene bears further investigation.

HHMI BioInteractive Animation

Using p53 to Fight Cancer

image

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For the present, the diagrams in Figure 18.24 and Figure 18.25 are an accurate view of how mutations can contribute to cancer, but we still don’t know exactly how a particular cell becomes a cancer cell. Recent studies have shown, for instance, that DNA methylation and histone modification patterns in normal cells differ from those in cancer cells and that miRNAs probably participate in cancer development. There is still a lot to learn, and you and your classmates may be the ones to make important discoveries about cancer biology.

The Multistep Model of Cancer Development

More than one somatic mutation or epigenetic change is generally needed to produce all the changes characteristic of a full-fledged cancer cell. This may help explain why the incidence of cancer increases greatly with age. If cancer results from an accumulation of mutations that occur throughout life, then the longer we live, the more likely we are to develop cancer.

The model of a multistep path to cancer is well supported by studies of one of the best-understood types of human cancer: colorectal cancer, which affects the colon and/or rectum. About 140,000 new cases of colorectal cancer are diagnosed each year in the United States, and the disease causes 50,000 deaths per year. Like most cancers, colorectal cancer develops gradually **(Figure 18.26)**. The first sign is often a polyp, a small, benign growth in the colon lining. The cells of the polyp look normal, although they divide unusually frequently. The tumor grows and may eventually become malignant, invading other tissues. The development of a malignant tumor is paralleled by a gradual accumulation of mutations that convert proto-oncogenes to oncogenes and knock out tumor-suppressor genes. A *ras* oncogene and a mutated *p53* tumor-suppressor gene are often involved.

Figure 18.26

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A multistep model for the development of colorectal cancer.
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Diagram modeling development of colorectal cancer from the loss of the tumor suppressor gene to the development of a malignant tumor (carcinoma).

This type of cancer is one of the best understood. Changes in a tumor parallel a series of genetic changes, including mutations affecting several tumor-suppressor genes (such as p53) and the ras proto-oncogene. Mutations of tumor-suppressor genes often entail loss (deletion) of the gene. APC stands for adenomatous polyposis coli, and SMAD4 is a gene involved in signaling that results in apoptosis.

Man first has normal colon epithelial cells in the colon wall. 1. Loss of tumor-suppressor gene (A P C or other) leads to a small benign growth (polyp, drawn here as a small mass of pink cells on the colon wall) which leads to, 2, Activation of ras oncogene and 32 loss of tumor-suppressor gene S M A D 4 and the small polyp becomes a larger benign growth (adenoma) and then 42 loss of tumor-suppressor gene p 53 and 5, additional mutations leads to a malignant tumor (carcinoma); drawn here as a large mass of pink cells that invade into the colon wall.

About half a dozen changes must occur at the DNA level for a cell to become fully cancerous. These changes usually include the appearance of at least one active oncogene and the mutation or loss of several tumor-

suppressor genes. Furthermore, since mutant tumor-suppressor alleles are usually recessive, in most cases mutations must knock out *both* alleles in a cell's genome to block tumor suppression. (Most oncogenes, on the other hand, behave as dominant alleles.)

Since we understand the progression of this type of cancer, routine screenings (colonoscopies, for example) are recommended to identify and remove any suspicious polyps. The colorectal cancer mortality rate has been declining for the past 20 years due to increased screening and improved treatments. Treatments for other cancers have improved as well. Advances in the sequencing of DNA and mRNA allow medical researchers to compare the genes expressed by different types of tumors and by the same type in different people. These comparisons have led to personalized treatments based on the molecular characteristics of a person's tumor; **Figure 18.27** shows how this approach has been applied to breast cancer.

Figure 18.27

AP Make Connections: Genomics, Cell Signaling, and Cancer

Genomics, cell signaling, and cancer. Normal Breast Cells in Milk Duct: E R alpha plus, PR plus, H E R 2 plus. Breast cancer subtypes.

Modern medicine that melds genome-wide molecular studies with cell-signaling research is transforming the treatment of many diseases, such as breast cancer. Using microarray analysis (see figure 20.13) and other techniques, researchers measured the relative levels of m R N A transcripts for every gene in hundreds of breast cancer tumor samples. They identified four major subtypes of breast cancer that differ in their expression of three signal receptors involved in regulating cell growth and division:

estrogen receptor alpha (E R alpha)

progesterone receptor (P R)

H E R 2, a type of receptor called a receptor tyrosine kinase (see figure 11.8)

(E R alpha and P R are steroid receptors; see figure 11.9.) The absence or excess expression of these receptors can cause aberrant cell signaling, leading in some cases to inappropriate cell division, which may contribute to cancer (see figure 18.24). Photo shows a research scientist at a computer screen, examining D N A sequencing data from breast cancer samples. Normal Breast Cells in Milk Duct: In a normal breast cell, the three signal receptors are at normal levels (indicated here by plus signs): E R alpha +, P R +, and H E R 2+. Drawing shows normal cell adjacent to support cells and surrounded by extracellular matrix. Duct interior towards center of group of cells drawn. Normal cell diagrammed showing: nucleus with D N A and two estrogen receptors alpha (E R alpha). In the cytoplasm, drawn are two progesterone receptors (P R) and four H E R 2s (a receptor tyrosine kinase), which span the cell membrane.

" Each breast cancer subtype is characterized by the overexpression (indicated by 2 plus signs or 3 plus signs) or absence (shown by a minus sign) of three signal receptors: E R alpha, PR, and H E R 2. Breast cancer treatments are becoming more effective because they can be tailored to the specific cancer subtype.

Breast cancer subtypes: Luminal A: E R alpha 3 plus, PR 2 plus, H E R 2 minus, 40 percent of breast cancers, best prognosis. Breast cell shown has nucleus with 5 E R alphas in the nucleus, 5 PRs in the cytoplasm, and no H E R 2s.

Breast cancer subtypes: Luminal B: E R alpha two plus, PR two plus, H E R 2 minus (shown, some H E R 2 plus), 15-20% of breast cancers, poorer prognosis than luminal A subtype. Breast cell shown has 3 E R alphas in the nucleus, 5 PRs in the cytoplasm, and no H E R 2s. Both luminal A and luminal B subtypes overexpress E R alpha (luminal A more than luminal B) and PR, and usually lack expression of H E R 2. Both can be treated with drugs that target E R alpha and inactivate it, the most well-known drug being tamoxifen. These subtypes can also be treated with drugs that inhibit estrogen synthesis.

Breast cancer subtypes: Basal-like. E R alpha minus, PR minus, H E R 2 minus, 15-20 percent of breast cancers, more aggressive; poorer prognosis than other subtypes. Breast cell shown has a longer, thinner shape than other breast cells and shows none of the elements diagramed in the other subtypes. The basal-like subtype is triple negative – it does not express E R alpha, PR, or H E R 2. It often has a mutation in the tumor-suppressor gene BRCA1 (see concept 18.5). Treatments that target ER, PR, or H E R 2 are not effective, but new treatments are being developed. Currently, patients are treated with cytotoxic chemotherapy, which selectively kills fast-growing cells. The H E R 2 subtype overexpresses H E R 2. Because it does not express either E R alpha or PR at normal levels, the cells are unresponsive to therapies that target those to receptors. However, patients with the H E R 2 subtype can be treated with Herceptin, an antibody protein that inactivates H E R 2 (see concept 12.3). Drawing shows breast cancer subtype H E R 2. E R alpha minus, PR minus, H E R 2 plus, 10-15 percent of breast cancers, poorer prognosis than luminal A subtype. Breast cell shown has no E R alpha in nucleus, no PRs in cytoplasm, and 14 H E R 2s in the cell membrane. Series of five drawings summarize the development of H E R 2 breast cancer as follows:

- 1. Signaling molecules (such as a growth factor) bind to H E R 2 receptor monomers (single receptor proteins).
- 2. Binding of signaling molecules causes two receptor monomers to associate closely with each other, forming a dimer.
- 3. Formation of a dimer activates each monomer.
- 4. Each monomer adds phosphate from ATP to the other monomer, triggering a signal

transduction pathway.

- 5. The signal is transduced through the cell, which leads to a cellular response – in this case, turning on genes that trigger cell division. H E R 2 cells have up to 100 times as many H E R 2 receptors as normal cells, so they undergo uncontrolled cell division.

Two drawings summarize treatment with Herceptin for the H E R 2 some type as follows:

- 1. The drug Herceptin binds to H E R 2 receptors in place of the usual signaling molecules.
- 2. In certain patients with H E R 2 subtype, signaling is blocked and excessive cell division does not occur.

MAKE CONNECTIONS *When researchers compared gene expression in normal breast cells and cells from breast cancers, they found that the genes showing the most significant differences in expression encoded signal receptors, as shown here. Given what you learned in*Chapters 11*,***12**, and this chapter, explain why this result is not surprising.*

*For suggested answer, see*Appendix A*.*

Breast cancer is the second most common form of cancer in the United States, and the first among women. Each year, this cancer strikes over 230,000 women (and some men) in the United States and kills 40,000 (450,000 worldwide). A major problem with understanding breast cancer is its heterogeneity: Tumors differ in significant ways. Identifying differences between types of breast cancer is expected to improve treatment and decrease the mortality rate. In 2012, the Cancer Genome Atlas Network, sponsored by the National Institutes of Health, published the results of a multi-team effort that used a genomics approach to profile subtypes of breast cancer based on their molecular signatures. Four major types of breast cancer were identified (Figure 18.27). It is now routine to screen for the presence of particular signaling receptors in any breast cancer tumors, and individuals with breast cancer, along with their physicians, can now make more informed decisions about their treatments.

Inherited Predisposition and Environmental Factors Contributing to Cancer

The fact that multiple genetic changes are required to produce a cancer cell helps explain the observation that cancers can run in families. An individual inheriting an oncogene or a mutant allele of a tumor-

suppressor gene is one step closer to accumulating the necessary mutations for cancer to develop than is an individual without any such mutations.

Geneticists are working to identify inherited cancer alleles so that predisposition to certain cancers can be detected early in life. About 15% of colorectal cancers, for example, involve inherited mutations. One syndrome, called *hereditary nonpolyposis colon cancer (HNPCC),* increases an individual's lifetime risk of colon cancer to 50–70%. HNPCC, also known as Lynch syndrome, is caused by an autosomal dominant allele of any one of a group of DNA repair genes, underscoring the importance of DNA repair systems. This syndrome is responsible for 2–5% of colon cancers. Other inherited mutations that cause colon cancer affect the tumor-suppressor gene called *adenomatous polyposis coli*, or *APC* (see Figure 18.26). This gene has multiple functions in the cell, including regulation of cell migration and adhesion. Even in patients with no family history of the disease, the *APC* gene is mutated in 60% of colorectal cancers. In these individuals, new mutations must have occurred in both *APC* alleles before the gene's function is lost. Currently, only 15% of colorectal cancers are associated with known inherited mutations, so researchers continue to try to identify "markers" that could predict the risk of developing this type of cancer.

Given the prevalence and significance of breast cancer, it is not surprising that it was one of the first cancers for which the role of inheritance was investigated. It turns out that for 5–10% of patients with breast cancer, there is evidence of a strong inherited predisposition. Geneticist Mary-Claire King began working on this problem in the mid-1970s. After 16 years of research, she convincingly demonstrated that mutations in one gene—BRCA1—were associated with increased susceptibility to breast cancer, a finding that flew in the face of medical opinion at the time. (*BRCA* stands for breast cancer.) Mutations in that gene or a gene called BRCA2 are found in at least half of inherited breast cancers, and tests using DNA sequencing can detect these mutations. A woman who inherits one mutant BRCA1 allele has a 60% probability of developing breast cancer before the age of 50, compared with only a 2% probability for an individual homozygous for the normal allele.

Interview

Interview with Mary-Claire King: Discovering breast cancer genes

Mary-Claire King.

BRCA1 and BRCA2 are considered tumor-suppressor genes because their wild-type alleles protect against breast cancer and their mutant alleles are recessive. (Mutations in BRCA1 are commonly found in the genomes of cells from basal-like breast cancers; see Figure 18.27.) The BRCA1 and BRCA2 proteins both appear to function in the cell's DNA damage repair pathway. More is known about BRCA2: Along with

another protein, it helps repair breaks that occur in both strands of DNA, a function crucial for maintaining undamaged DNA.

Because DNA breakage can contribute to cancer, it makes sense that the risk of cancer can be lowered by minimizing exposure to DNA-damaging agents, such as the ultraviolet radiation in sunlight and chemicals found in tobacco products. Novel genomics-based analyses of specific cancers, such as the approach described in Figure 18.27, are contributing to both early diagnosis and development of treatments that interfere with expression of key genes in tumors. Ultimately, such approaches are expected to lower the death rate from cancer.

The Role of Viruses in Cancer

The study of genes associated with cancer, inherited or not, increases our basic understanding of how disruption of normal gene regulation can result in this disease. In addition to the mutations and other genetic alterations described in this section, a number of *tumor viruses* can cause cancer in various animals, including humans. In fact, one of the earliest breakthroughs in understanding cancer came in 1911, when Peyton Rous, an American pathologist, discovered a virus that causes cancer in chickens. Also, the Epstein-Barr virus, which causes infectious mononucleosis, has been linked to several types of cancer in humans, notably Burkitt's lymphoma. Papillomaviruses cause cancer of the cervix, and a virus called HTLV-1 causes a type of adult leukemia. Viruses play a role in about 15% of the cases of human cancer.

Viruses may at first seem very different from mutations as a cause of cancer. However, we now know that viruses can interfere with gene regulation in several ways if they integrate their genetic material into the DNA of a cell. Viral integration may donate an oncogene to the cell, disrupt a tumor-suppressor gene, or convert a proto-oncogene to an oncogene. Some viruses produce proteins that inactivate p53 and other tumor-suppressor proteins, making the cell more prone to becoming cancerous. Viruses are powerful biological agents; you'll learn more about them in Chapter 19.

Animation

Causes of Cancer

image

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